

Executive Summary

1. Correlation between order status and discount levels.

Cancelled orders exhibit the highest average discount at 12.37%, followed by delivered orders at 12.19%. Conversely, returned orders show the lowest average discount among these categories, standing at 11.48%.

Conclusion: The variance is marginal; however, cancelled orders received the highest average discount. This suggests that customers may more frequently cancel orders made at highly attractive price points, or that deep discounting is not a sufficient incentive to ensure transaction completion.

"2. Correlation Analysis: The impact of discounts on purchase value.

The scatter plot illustrates the relationship between the discount percentage (X-axis) and the purchase value (Y-axis). The trend line is nearly horizontal with a very slight negative slope.

Conclusion: There is a very weak trend suggesting that orders with higher discount percentages have a slightly lower purchase value. Overall, discounts have virtually no impact on the total order value. High-value orders occur consistently across the entire discount range (from 5% to 20%), and the same pattern is observed for low-value orders.

3. Average Order Value (AOV) by order status.

Cancelled and Returned orders exhibit the highest average values, at 11,738.74 and 11,725.87 respectively. In contrast, Delivered orders have a lower average value of 11,104.47, while Pending orders show the lowest at 9,468.14.

Conclusion: Customers are more likely to cancel or return high-value orders. This trend may be attributed to heightened expectations for expensive items, more cautious decision-making ('buyer's remorse'), or perceived financial risks associated with larger transactions.

4. Problematic Order Analysis by Device Type and Payment Method.

Conclusion: There is a significant technical or operational friction associated with Desktop-based transactions, particularly those using Bank Transfer and PayPal. A failure rate exceeding 50% across all top combinations indicates that the issue is not isolated but systemic. The peak in Bank Transfer issues suggests that the manual steps or external redirects required for this method on Desktop may be leading to high abandonment or processing errors.

Conclusion: The high proportion of problematic orders—particularly cancellations and returns—associated with Desktop usage and specific payment methods (Bank Transfer, PayPal) may point to Desktop-specific UX friction. Alternatively, it suggests that these payment methods are frequently used for purchases that customers are more inclined to reconsider or return.

5. Average Order Value (AOV) by Country.

Conclusion: There is a clear trade-off between order volume and transaction value across key markets. The USA is the most high-value market per customer, suggesting a premium audience. Germany, conversely, is the primary driver of sales volume, though with a lower price point per order. This divergence indicates that Germany is a high-penetration market, whereas the USA represents a high-margin opportunity.

Conclusion: The USA is the most premium market in terms of Average Order Value (AOV). This region should be prioritized as the primary target market for high-end, luxury, or high-margin product categories.

"6. Average number of orders per user by country.

Conclusion: The business currently relies entirely on new customer acquisition, as the repeat purchase rate is zero across all markets. There is no evidence of customer retention or loyalty-driven revenue in the current dataset. Every transaction represents a unique user, indicating that the brand has yet to establish a recurring relationship with its audience.

Conclusion: The vast majority of users in this dataset have placed only a single order. This is a critical metric indicating a lack of repeat purchases and near-zero customer loyalty. It is imperative to immediately pivot toward retention strategies, including CRM optimization, loyalty programs, and targeted remarketing campaigns for existing customers.

7. Percentage of Delivered Orders by Category and Country.

Conclusion: The operation faces a systemic delivery crisis. With the best-performing segment failing to reach even a 50% success rate, the business is operating with extreme inefficiency. This suggests that the primary bottleneck is not marketing or customer interest, but a failure in the post-order fulfillment chain—potentially due to logistics, payment processing, or supply chain instability.

Conclusion: The low delivery percentage across all combinations confirms a systemic issue with order fulfillment, cancellations, returns, and pending statuses. This represents a severe operational risk to the business, as more than half of all generated demand fails to convert into completed sales.

8. Combined Percentage of Problematic Orders (Cancellations & Returns) by Category and Country.

The highest combined problematic order rate is 73.08% for the 'Home' category in the USA, driven by a critically high return rate of 46.15%. Additionally, 'Books' in Germany show a severe failure rate of 67.86%, with 38.57% of orders being cancelled before fulfillment.

Conclusion: An urgent investigation is required into the root causes of the excessive return rate for the 'Home' category in the USA and the high cancellation rate for 'Books' in Germany. These two segments currently represent the most significant drain on business efficiency and profitability.

9. RFM Analysis for VIP Customer Identification.

Conclusion: The VIP segment is extremely fragile. The brand is successful at attracting 'High-Monetary' individuals, but fails completely at the 'Frequency' component of RFM. A Recency gap of up to 299 days for top spenders suggests a 'leaky bucket' syndrome: once a premium customer completes (or attempts) a purchase, they are effectively lost to the brand.

Conclusion: The RFM analysis clearly demonstrates that the company lacks a genuine 'VIP' or loyal customer base. All retention efforts must be urgently redirected toward converting these high-value one-time buyers into recurring customers.

10. Analysis of Sales Volumes by Category, Country, Time, and Order Distribution.

Conclusion: The business is heavily reliant on the 'Books' category, which generates nearly a quarter of all sales. However, considering our previous findings, this leadership is fragile. Since Books suffer from high cancellation rates in key markets like Germany, and the 'Home' category (16.5%) is plagued by returns in the USA, more than 40% of the company's total sales volume is currently under high operational risk.

Conclusion: India is the primary engine of the company's sales volume, contributing over a quarter of all transactions. This market, combined with the UK, forms the core of the business's operational stability. However, the USA — while being the most 'expensive' market per customer — ranks only third in volume and suffers from the highest attrition rates. The business currently stands on a 'Volume vs. Value' divide: India provides the scale, while the USA provides the potential margin, though currently hindered by systemic fulfillment issues.

Conclusion: The business is in a 'stagnant equilibrium.' The lack of a growth trend, coupled with zero repeat purchases, confirms that the company is failing to build equity. Sales spikes are purely seasonal and temporary, while the dips in August and September indicate a high vulnerability to seasonal fluctuations in consumer activity.

Conclusion: The business operates on a 'long-tail' model. Most revenue is generated by a high volume of low-cost items, while the high-value segment (>16,000) is unstable. The fact that the median is pulled toward the lower end suggests that the 'premium' image of the brand is not yet reflected in the behavior of the mass customer. High-value orders are outliers, not the norm.

Executive Summary: The core business revolves around Books and Toys. Given the high rate of problematic orders (returns) in the 'Home' category in the USA, its significant 16.5% share of total sales volume makes this operational issue critical. While the USA offers the highest average order value (AOV), India generates the largest overall sales volume. This indicates that India is a high-frequency/lower-ticket market, whereas the USA is a premium/lower-frequency market. Sales exhibit relative stability throughout the year, allowing for strategic promotional planning around identified peak months.

Recommendations

1. Increasing Loyalty and Repeat Sales.

Establish a Post-First-Purchase Program (Automated campaign for one-time buyers, offering personalized discounts or bonuses for their second purchase). Revise the RFM strategy (Focus on stimulating 'Monetary Value' for those who made a large but one-time order; provide special offers and premium service for this segment).

2. Optimization of Problematic Order Management.

Conduct a deep-dive analysis into the causes of the 73.08% problematic order rate for the "Home" category in the USA. Specifically, audit product quality, packaging standards, and logistics efficiency. Investigate the root causes of the high cancellation rate (38.57%) for "Books" in Germany. Potential issues to explore include stock availability, delivery lead times, or friction within the checkout process. Audit the User Experience (UX) for "Bank Transfer" payments, particularly on Desktop. High failure rates suggest technical or usability barriers during the final stages of the transaction.

"3. Adjustment of Pricing and Promotional Policies.

Discounting for Retention: Instead of high upfront discounts, offer "Loyalty Credits" valid for the next purchase to convert one-time buyers into repeat customers.

Premium Service Tier: Introduce a "White-Glove Delivery" option for the USA market, including assembly or personal inspection, to justify the 12k+ price point and ensure customer satisfaction.

4. Geographical Strategy.

India (Volume Market): As India represents the largest sales volume at 25.2%, the primary focus must be on reducing logistics costs and optimizing operations to support a high frequency of low-ticket orders.

USA (Value Market): Implement enhanced quality control and premium delivery services for the USA to align with the high Average Order Value (AOV) and meet premium customer expectations.